

Chapter 29 — Sexually Transmitted Diseases

EPIDEMIOLOGY OF SEXUALLY TRANSMITTED DISEASES

- **“Hidden epidemics”**: a 1997 Institute of Medicine (IOM) report described sexually transmitted diseases (STDs) as hidden epidemics of major health and economic consequence, under-recognised by the public and by professionals; **this remains true two decades later**.
- **Resurgence**: syphilis and gonorrhoea, previously declining, are rising again. Against 2016 Centers for Disease Control and Prevention (CDC) data, gonorrhoea rose 45% and syphilis 52%, but **congenital syphilis rose 235%** — the steepest rise.
- **Burden**: about 67.6 million prevalent infections in the United States (US) in 2018 — **roughly one adult in five has an STD**; direct treatment costs nearly **\$16 billion a year**.

TABLE 29.1 SEXUALLY TRANSMITTED DISEASE CASES, 2021

Infection	Cases reported	Rate per 100,000	Increase from 2020
Chlamydia	1,644,416	495.5	3.9%
Gonorrhoea	710,151	214.0	4.6%
Syphilis	176,713	53.2	28.6%

From CDC: Sexually Transmitted Disease Surveillance, 2023.

- **What the table tests**: **chlamydia is the most frequent** reportable STD; **syphilis is the fastest rising**. The trap is to equate the two.

Clinical Prevention Guidance

- **Five major strategies**: (1) accurate risk assessment, education and counselling; (2) pre-exposure vaccination against vaccine-preventable STDs; (3) identification of symptomatic and asymptomatic infected persons; (4) effective diagnosis, treatment, counselling and follow-up; (5) evaluation, treatment and counselling of sex partners.
- **The first step is identifying the at-risk population**. The **“Five P’s”** give a gender-inclusive and sexual orientation-inclusive history: **partners, practices, protection against STDs, past STD history, pregnancy intention**.
- **Risk factors**: more lifetime partners, condomless sex, substance misuse, risky partners.
- **Men who have sex with men (MSM)**: human immunodeficiency virus (HIV) and bacterial STDs fell in this group in the mid-1990s, but syphilis, gonorrhoea and chlamydia are rising again; **MSM account for about half of male primary and secondary syphilis**.

Screening Recommendations

- **Risk-tailored**: a 2014 multi-organisation report advised assessing **behavioural and biological risk** and tailoring counselling and screening to it. The nucleic acid amplification test (NAAT) is the standard screening assay.

SCREENING BY POPULATION — CDC 2021

Population	Chlamydia and gonorrhoea	Syphilis	HIV	Hepatitis
Young or at-risk females	Annually if sexually active and ≤24 y; ≥25 y if risk factors	—	—	—

Population	Chlamydia and gonorrhoea	Syphilis	HIV	Hepatitis
Pregnant patients	First prenatal visit if <25 y or at increased risk	Serology for all at first visit	All at first visit, even if tested before; retest in third trimester if high risk	Hepatitis C virus (HCV) at first visit if at risk
Adolescents	Annually, all sexually active females <25 y	Routinely in young MSM and pregnant adolescents	Discuss and offer to all	—
Correctional facilities	Females ≤35 y, males <30 y	Universal screening guided by local and institutional prevalence of early syphilis	All	Hepatitis B virus (HBV) and HCV, all
MSM	Urethral (insertive, past year; urine NAAT) and rectal (receptive anal, past year; rectal NAAT) regardless of condom use; pharyngeal for gonorrhoea only (receptive oral)	Serology — distinguishes untreated, partially treated, slow serological response and serofast states	If status unknown or negative and patient or partner has had >1 partner since last test	HBV: surface antigen (HBsAg), core antibody, surface antibody
Transgender and gender-diverse	By current anatomy and behaviour ; annually if cervix present and <25 y	If HIV-positive with cisgender male or transgender female partners	Review and offer to all	HCV in the same HIV-positive group

- **Pattern:** screening follows **anatomy and exposure site, not gender label** — hence rectal and pharyngeal sites in MSM, and cervical rules applied to transgender males and non-binary persons with a cervix.
- **Transgender specifics:** after **vaginoplasty**, screen all exposed sites; after **metoidioplasty without vaginectomy**, include a cervical swab; HIV-positive persons with cisgender male or transgender female partners need syphilis, HCV, and urogenital plus extragenital NAAT for gonorrhoea and chlamydia.
- **Trap:** **pharyngeal chlamydia testing is not recommended**; pharyngeal gonorrhoea testing is.

Reporting Sexually Transmitted Diseases

- **Six conditions are reportable in every US state:** syphilis (including congenital), gonorrhoea, chlamydia, chancroid, HIV infection and acquired immunodeficiency syndrome (AIDS). Other STDs vary by state; local health departments list them.

Prevention Methods

- **Methods:** pre-exposure vaccination, abstinence and fewer partners, male and female condoms, cervical diaphragms, topical microbicides and spermicides.
- **Male condoms used consistently are highly effective against HIV:** in heterosexual serodiscordant couples (one partner infected), consistent use made the negative partner about 80% less likely to acquire HIV.
- **Other STDs:** condoms reduce risk and, by limiting lower genital tract inflammation, may reduce pelvic inflammatory disease (PID). Against human papillomavirus (HPV), herpes, syphilis and chancroid they protect **only where the site of exposure is covered**.
- **Quality:** regulated by the Food and Drug Administration (FDA) as medical devices with random sampling; **breakage about 2 per 100**, slightly higher in anal intercourse.

CONDOM MATERIALS COMPARED

Material	STD and HIV protection	Compatible lubricant	Note
Latex	Reference standard	Water- or silicone-based only — oil weakens latex	Discard at expiry or >5 y after manufacture
Polyurethane	Comparable to latex, including pregnancy	Oil- or water-based	—
Other synthetics	Not extensively studied	—	—
Natural membrane ("lambskin", lamb caecum)	No protection against HIV or HBV	—	Pores up to 1500 nm: block sperm, pass viruses

Concept — *prevention is risk-stratified: the Five P's history defines risk, risk defines which sites are screened and how often, and each barrier method protects only the surface it physically covers.*

URETHRITIS

- **Presentation:** urethral discharge, dysuria, urethral stinging or itching, penile tip irritation — but **often asymptomatic**.

Etiology

- **Complications** of the estimated 2.8 million annual cases: acute epididymitis, orchitis, prostatitis.
- **Gram-negative intracellular diplococci** → **presumed *Neisseria gonorrhoeae***.
- **Nongonococcal urethritis (NGU):** *Chlamydia trachomatis* is the **commonest identifiable cause**; *Mycoplasma genitalium* is the **second commonest** and the **leading identified cause of persistent or recurrent urethritis**.
- **Less common:** *Trichomonas vaginalis*, herpes simplex virus (HSV), *Gardnerella* (formerly *Haemophilus*) *vaginalis*, *Neisseria meningitidis*, adenovirus.

Diagnosis

- **Threshold lowered:** the old Gram stain criterion of **≥5 white blood cells (WBC) per high-power field (hpf)** missed a significant proportion of chlamydia, gonorrhoea and *M. genitalium* → CDC now uses **≥2 WBC per oil immersion field**.
- **Gram stain sensitivity** depends on operator and collection device; yield rises **swab → loop → spatula**.
- **Urethritis is documented by any one of:** mucoid, mucopurulent or purulent discharge; Gram stain **≥2 WBC per oil immersion field**; positive leucocyte esterase on first-void urine; **≥10 WBC/hpf in first-void urine**.
- **No point-of-care (POC) Gram stain → NAAT for all males; urine is the preferred male specimen.** NAAT is also indicated when **≥2 WBC per oil immersion field** are seen without intracellular gram-negative or purple diplococci.

Gonococcal Urethritis

- **Rising:** reported cases in 2020 were more than double those of 2009. Gram-negative diplococcus; **incubation 3–14 days**.
- **Sex asymmetry: males are usually symptomatic** (urethritis, epididymitis, prostatitis, proctitis; rarely disseminated gonococcal infection); **females are usually asymptomatic** and present with sequelae — PID, tubal scarring, infertility, ectopic pregnancy, chronic pelvic pain.

Diagnosis of Gonococcal Urethritis

- **Diagnostic in symptomatic males:** Gram stain showing polymorphonuclear leucocytes with intracellular gram-negative diplococci. **A negative Gram stain does not exclude gonorrhoea** (low sensitivity).

- **NAAT is preferred** for gonorrhoea and chlamydia (higher sensitivity); POC NAAT (GeneXpert) exists. NAAT runs on first-catch urine or urethral swab — **first-void urine is preferred**; culture requires a urethral swab. FDA-approved NAATs cover urine, endocervical, vaginal, male urethral, rectal and pharyngeal specimens; follow each manufacturer's insert.
- **Culture is still indicated** for suspected treatment failure and resistance monitoring. In children, use culture or only Clinical Laboratory Improvement Amendments (CLIA)-validated, FDA-cleared NAATs, given the implications of the diagnosis.
- **Sampling order:** when several specimens come from one site, **take the gonococcal culture first** to maximise yield.

Treatment of Gonococcal Urethritis

- **Why it is complex:** accumulating **antimicrobial resistance** and less culture (so fewer susceptibility data). The CDC Gonococcal Isolate Surveillance Project (GISP), established 1986, monitors susceptibility.

GONORRHOEA TREATMENT — CDC 2021 VS EUROPEAN 2020

Situation	CDC 2021	European 2020
First line	Ceftriaxone 500 mg intramuscular (IM) single dose (<150 kg); 1 g if ≥150 kg — monotherapy	Ceftriaxone 1 g + azithromycin 2 g; ceftriaxone 1 g alone only in well-controlled settings
Ceftriaxone unavailable	Gentamicin 240 mg IM + azithromycin 2 g orally; or cefixime 800 mg orally, single doses	—
IgE-mediated penicillin allergy strongly suspected, or cephalosporin allergy	Gentamicin 240 mg IM + azithromycin 2 g orally	—
Ciprofloxacin	Only if asymptomatic and gyrase A (gyrA) testing available	—
Test of cure	Pharyngeal infection only, 7–14 days after treatment	All cases

- **CDC has abandoned dual ceftriaxone–azithromycin therapy;** Europe retains it.
- **Extragenital sites:** failure rates of non-ceftriaxone regimens can be substantial.
- **Penicillin allergy:** penicillins and cephalosporins share a β -lactam ring. With confirmed immunoglobulin E (IgE)-mediated penicillin allergy, cross-reactivity with third-generation cephalosporins is **<1%**, yet CDC advises against β -lactams for such patients in the ambulatory setting. **Third- and fourth-generation cephalosporins are safe if there has been no IgE-mediated reaction (anaphylaxis, urticaria) to penicillin in the preceding 10 years.**
- **If chlamydia has not been excluded,** add antichlamydial therapy.
- **Treatment failure** = no symptom resolution within 3–5 days without sexual contact, and/or a positive test of cure without reported contact → **culture with simultaneous NAAT and susceptibility testing** (Etest or agar dilution).
- **Aftercare:** test for chlamydia, syphilis and HIV; **abstain for 7 days after treatment** and until partners are treated; refer **partners from the 60 days before symptom onset** for testing and presumptive treatment; **retest at 3 months** regardless of partner treatment (high recurrence).

Nongonococcal Urethritis

- **Proportions:** gonorrhoea causes 5–20% of urethritis; the remainder is NGU. *C. trachomatis* leads (more so in younger males), *M. genitalium* follows. **Many males never receive a definitive diagnosis — no pathogen is found in up to half even with extensive testing.**

- **Recurrent or persistent symptoms after empiric treatment → test for *M. genitalium*.**
- **Minor pathogens:** *Ureaplasma urealyticum* (weak association, young males with few partners; consider when nothing else is found); *T. vaginalis* (consider in high-prevalence areas or after empiric failure); also *N. meningitidis*, *Haemophilus influenzae*, *Corynebacterium*, HSV-1, HSV-2, adenovirus.
- **Treatment:** pathogen-directed when possible, otherwise presumptive. **Doxycycline 100 mg orally twice daily for 7 days;** alternatives azithromycin 1 g single dose, or 500 mg then 250 mg daily for 4 days. Doxycycline covers chlamydia well and has some *M. genitalium* activity. Consider concurrent gonorrhoea treatment in at-risk patients or high-prevalence areas.

Chlamydia

- **Most frequently reported infectious disease in the US,** peaking under 24 years; incubation 3–14 days.
- **Male sequelae:** epididymitis, reactive arthritis (formerly Reiter syndrome). Treating males also protects female partners from ascending infection — tubal scarring, PID, pelvic pain, infertility.
- **First-catch urine NAAT is the most sensitive male test and is recommended.**
- **Treatment: doxycycline 100 mg twice daily for 7 days;** azithromycin regimens are alternatives. **This changed from 2015,** when doxycycline and azithromycin were equal first-line options; the 2015 European guideline is under review. Consider concurrent gonorrhoea treatment.
- **Aftercare:** abstain for 7 days after single-dose therapy, or until the 7-day course is complete and symptoms have resolved; test for gonorrhoea, HIV and syphilis; **no test of cure** (retesting within 3 weeks may detect residual nucleic acid) but **retest at 3 months;** partners from the preceding 60 days.

Mycoplasma genitalium

- **Share of disease:** first identified in the early 1980s; causes a substantial minority of NGU, a larger share of non-chlamydial NGU, and the **largest identified share of persistent or recurrent urethritis.**
- **Transmission:** mainly genital–genital mucosal contact; penile–anal transmission documented; oral–genital unlikely (low oropharyngeal concentration).
- **Diagnosis: no cell wall → cannot be Gram stained;** culture takes up to 6 months → **NAAT is the only clinically useful test** (FDA-approved on urine, urethral, meatal, endocervical and vaginal swabs). Test males with recurrent NGU.
- **Resistance: macrolide resistance is widespread and rising; resistance-guided therapy** achieves cure in over 90%. European 2021 guidance: every positive NAAT is followed by a macrolide-resistance assay.

TABLE 29.2 TREATMENT RECOMMENDATIONS FOR MYCOPLASMA GENITALIUM

Resistance testing	Result	Regimen
Available	Macrolide sensitive	Doxycycline 100 mg twice daily 7 days → azithromycin 1 g, then 500 mg daily for 3 further days
Available	Macrolide resistant	Doxycycline 100 mg twice daily 7 days → moxifloxacin 400 mg daily 7 days
Not available	NAAT positive	Doxycycline 100 mg twice daily 7 days → moxifloxacin 400 mg daily 7 days

Data from Workowski et al., CDC 2021.

- **Pattern:** every regimen **starts with doxycycline;** macrolide susceptibility picks the second drug. **Without testing, resistance is assumed → moxifloxacin.**
- **Trap:** single-dose azithromycin is no longer adequate for *M. genitalium*.
- **Aftercare:** abstain until treatment is complete and symptoms resolve; partners managed as for NGU (preceding 60 days); screen for chlamydia, gonorrhoea, syphilis, HIV and *T. vaginalis* where appropriate.

Trichomonas vaginalis

- **Flagellated parasite** of the male urethra and the female urethra, vagina and vulva; mostly minimal or no symptoms; **about 1.5-fold increased HIV acquisition**.
- **Testing in males is limited:** no FDA-cleared rapid test; NAATs are replacing wet mount and culture, but many are not cleared for males. **Not part of the initial NGU work-up** (low prevalence) — test male partners of infected females and males in high-prevalence areas.
- **Treatment (male): metronidazole 2 g orally once, or tinidazole 2 g orally once.**

Persistent or Recurrent Nongonococcal Urethritis

- **Three causes:** resistance, non-adherence, re-exposure to an untreated partner. Think of **resistant M. genitalium** and **T. vaginalis**.
- **Urological evaluation rarely finds a cause;** check for pain elsewhere in the pelvis suggesting chronic pelvic pain syndrome rather than localised urethritis.

URETHRITIS PATHOGENS — MANAGEMENT COMPARED

	Gonorrhoea	Chlamydia	M. genitalium	T. vaginalis (male)
Test	NAAT, first-void urine; culture for failure	First-catch urine NAAT	NAAT only	NAAT (few cleared for males)
First line	Ceftriaxone 500 mg IM	Doxycycline 7 days	Doxycycline → azithromycin or moxifloxacin	Metronidazole 2 g once
Test of cure	Pharyngeal only	No	Not stated	Not stated
Retest at 3 months	Yes	Yes	Not stated	Not stated
Abstinence	7 days and partners treated	7 days or course complete	Until treated and symptom-free	—
Partner window	60 days	60 days	60 days	All partners

Concept — urethritis management is shaped by two pressures: **diagnostic sensitivity** (the lowered WBC threshold and the move to NAAT) and **antimicrobial resistance** (higher-dose ceftriaxone monotherapy, culture for failures, and resistance-guided *M. genitalium* therapy).

EPIDIDYMITIS

- **Definitions:** **acute** = pain, swelling and inflammation of the epididymis for **<6 weeks**, sometimes with orchitis (epididymo-orchitis); **chronic** = scrotal, testicular or epididymal discomfort for **≥6 weeks**, infectious (commonest *Mycobacterium tuberculosis*) or non-infectious. Full discussion in Chapter 27.
- **Keep testicular torsion in mind** — the presentations overlap. Concurrent urethritis is common but often asymptomatic.

LIKELY ORGANISM BY PATIENT GROUP

Patient	Organisms
Younger males	<i>C. trachomatis</i> , <i>N. gonorrhoeae</i>
Insertive partner in anal intercourse	Above plus enteric organisms (<i>Escherichia coli</i> , <i>Pseudomonas</i>)
>35 y without insertive anal intercourse	Uropathogens (<i>E. coli</i>) from bladder outlet obstruction, urinary tract surgery or instrumentation

Diagnosis of Epididymitis

- **Clinical:** unilateral testicular pain and tenderness, reactive hydrocele, palpable epididymal swelling.
- **Ultrasonography is not used to diagnose epididymitis** — a negative scan would not change management; useful only when **torsion is suspected**.
- **Requires one of:** Gram or methylene blue stain of urethral secretion with ≥ 2 WBC per oil immersion field; positive leucocyte esterase on first-void urine; ≥ 10 WBC/hpf on spun first-void urine.
- **All suspected acute cases → NAAT for chlamydia and gonorrhoea.** Urine culture in sexually transmitted enteric infection and in older males with bacteriuria.

Treatment of Epididymitis

- **Start empiric therapy before results.**

EMPIRIC REGIMENS FOR ACUTE EPIDIDYMITIS — CDC

Likely cause	Regimen
Chlamydia or gonorrhoea	Ceftriaxone 500 mg IM once (1 g if ≥ 150 kg) + doxycycline 100 mg twice daily 10 days
Chlamydia, gonorrhoea and enterics (insertive anal sex)	Ceftriaxone 500 mg IM once + levofloxacin 500 mg daily 10 days
Enterics only, gonorrhoea excluded	Levofloxacin 500 mg daily 10 days

- **Duration is 10 days**, not the 7 days of uncomplicated urethritis.
- **European 2016 guidance** matches for chlamydial or gonococcal disease and adds **ofloxacin 200 mg twice daily for 14 days** — a secondary option there and a recommended option for enteric epididymitis.
- **Aftercare:** abstain until patient and partners are treated and symptoms resolve; test for other STDs and HIV.

Concept — *age and sexual practice predict the organism and the organism dictates the drug: doxycycline for chlamydia, ceftriaxone for gonorrhoea, levofloxacin for enterics.*

GENITAL, ANAL, AND PERIANAL ULCERS

- **Commonest causes in young sexually active US patients: genital herpes (most common), then syphilis;** less often chancroid and donovanosis. Non-infectious: carcinoma, yeast, trauma, psoriasis, fixed drug eruption.
- **History and examination are inaccurate → every ulcer gets:** syphilis serology plus darkfield examination or polymerase chain reaction (PCR) if available; HSV culture or NAAT plus type-specific HSV serology; *Haemophilus ducreyi* NAAT or culture in endemic areas; HIV test if status unknown.
- **About a quarter remain undiagnosed; biopsy ulcers that fail initial therapy or look unusual.**

TABLE 29.3 GENITAL ULCER DISEASE

Disease	Lesion	Lymphadenopathy	Systemic symptoms
Primary syphilis	Painless , indurated, clean base, usually single	Non-tender, rubbery, non-suppurative, bilateral	None
Genital herpes	Painful shallow vesicles, usually multiple	Tender, bilateral inguinal	During primary infection
Chancroid	Tender papule → painful, undermined, purulent ulcer, single or multiple	Tender, regional, suppurative	None

Disease	Lesion	Lymphadenopathy	Systemic symptoms
Lymphogranuloma venereum (LGV)	Small painless vesicle or papule → ulcer	Painful, matted, large nodes with fistulous tracts	After genital lesion heals
Donovanosis (from chapter text)	Painless, beefy red , bleeds easily	Absent	—

- **What the table tests — pain plus nodal pattern:** painless ulcer with painless nodes = syphilis; painful vesicles = herpes; painful ulcer with suppurative nodes = chancroid; transient painless lesion then painful matted nodes = LGV; painless ulcer without nodes = donovanosis.
- **Trap:** lesions are frequently atypical, so the table guides suspicion but **tests, not appearance, make the diagnosis.**

Genital Herpes Simplex Virus Infections

- **Virology:** HSV-1 and HSV-2 are double-stranded DNA viruses. **Most genital herpes is HSV-2**, but **HSV-1 now leads initial genital herpes in high-income countries**, especially in females and in MSM and transgender females under 25 who have sex with males.
- **Unrecognised infection is the rule:** about one in six US adults is HSV-2 seropositive, but only a minority recognise genital herpes. Shedding is intermittent, occurring on about 10% of days even in asymptomatic HSV-2 infection. **Transmission is greatest during lesional recurrence or prodrome.**
- **HSV-2 increases HIV acquisition two- to threefold → test all patients with genital herpes for HIV.**

Pathophysiology of Genital Herpes

- **Sequence:** epithelial cells → nerve endings → **retrograde axonal transport to sacral ganglia** → latency → reactivation: virus travels down the axon to skin and mucosa → shedding, with ulcers or, more often, asymptotically.
- **Recurrence is the hallmark:** median **4 recurrences in year one after HSV-2 vs 1.3 after HSV-1**; HSV-2 recurrences then fall by a median of 2 a year, with wide variation.
- **Reactivation triggers:** local trauma including surgery, ultraviolet light, immunosuppression, fever.

Presentation of Genital Herpes

- **Classic course:** painful papule → vesicle → ulcer with local adenitis, **4–7 days after exposure**. Many lesions are atypical — fissures, furuncle-like lesions, vulvar erythema.
- **Primary HSV-1 and HSV-2 are clinically indistinguishable** → serological typing needed.
- **Complications:** aseptic meningitis; **autonomic dysfunction causing urinary retention.**
- **Recurrences** are usually preceded by a prodrome, then resemble the primary lesions.

Diagnosis of Genital Herpes

- **Type-specific confirmation is required** because prognosis and counselling differ between HSV-1 and HSV-2.
- **Cell culture and NAAT are preferred**, swabbing the ulcer base; unroof vesicles with a needle or scalpel. FDA-cleared HSV NAATs are highly sensitive and specific.
- **A negative culture or NAAT does not exclude HSV** — shedding is intermittent, especially without active lesions.
- **HSV PCR is the test of choice for central nervous system (CNS) and systemic infection; blood PCR is not used for genital herpes** unless dissemination is suspected.
- **Tzanck preparation is insensitive — do not rely on it.**

- **Type-specific serology** uses glycoprotein G2 (HSV-2) and glycoprotein G1 (HSV-1); repeat at 12 weeks if recent HSV-2 acquisition is suspected. **Useful when:** (1) recurrent or atypical symptoms with negative PCR or culture; (2) clinical diagnosis without laboratory confirmation; (3) partner has genital herpes. **Not for general population screening;** consider in higher-risk persons with genital symptoms.

Treatment of Genital Herpes

- **Every symptomatic first episode receives antiviral therapy**, because severe or prolonged illness can follow.
- **Recurrence:** nearly all symptomatic first-episode HSV-2 recurs; HSV-1 recurs less often.
- **Two strategies for recurrent disease: suppressive** — reduces recurrence frequency and **reduces HSV-2 transmission to partners; episodic** — shortens lesions but **must start within 1 day of lesion onset or during the prodrome**.
- **Counselling** of patients and partners is central (full topic list in CDC 2021).

TABLES 29.4–29.6 RECOMMENDED ANTIVIRAL REGIMENS FOR GENITAL HERPES

Drug	First clinical episode, 7–10 days (Table 29.4)	Suppressive, recurrent (Table 29.5)	Episodic, recurrent HSV-2 (Table 29.6)
Aciclovir	400 mg three times daily	400 mg twice daily	800 mg twice daily 5 days; or 800 mg three times daily 2 days
Valaciclovir	1 g twice daily	500 mg once daily; or 1 g once daily	500 mg twice daily 3 days; or 1 g once daily 5 days
Famciclovir	250 mg three times daily	250 mg twice daily	125 mg twice daily 5 days; or 1 g twice daily 1 day; or 500 mg once then 250 mg twice daily 2 days

Data from Workowski et al., CDC 2021.

- **Footnotes:** first-episode therapy may be extended if healing is incomplete at 10 days; aciclovir 200 mg five times daily (first episode) and 400 mg three times daily for 5 days (episodic) work but are **not recommended because of dosing frequency; valaciclovir 500 mg once daily may be less effective with frequent recurrences (≥10 a year)**.
- **Pattern:** first-episode courses are longest; suppressive doses are low and continuous; episodic courses are short and front-loaded.

Syphilis

- **Organism:** *Treponema pallidum*, a coiled spirochaete; systemic disease, acquired or congenital.
- **Epidemiology:** MSM are disproportionately affected, but **female primary and secondary syphilis rose steeply (147%) from 2016 to 2020** — the driver of congenital syphilis.
- **Transmission:** sexual (vaginal, anogenital, orogenital); rarely blood products, organ donation, occupational exposure. **Over half of sexual contacts of early syphilis acquire infection.**
- **HIV link: all primary or secondary syphilis → HIV test; offer pre-exposure prophylaxis (PrEP) if negative.** Syphilis is far more common in people with HIV and increases HIV shedding in plasma and genital tract. HIV can produce unusual serological responses — when clinical suspicion persists despite non-reactive or unclear serology, use biopsy or darkfield examination.

STAGES OF SYPHILIS

Stage	Timing	Features
Primary	Incubation 10–90 days ; heals untreated in 3–8 weeks	Chancre: painless, single, indurated, clean base, clear serous discharge; regional lymphadenopathy. Often atypical — multiple, deep, herpes-like

Stage	Timing	Features
Secondary	Bacteraemia 3–5 months after infection	Non-pruritic maculopapular rash including palms and soles ; mucosal lesions (vagina, anus); condylomata lata (large, raised, grey or white); fever, lymphadenopathy, alopecia, weight loss. Vasculitis in <10%: hepatitis, iritis, nephritis, headache, cranial nerve VIII ; neurological sequelae managed as early neurosyphilis
Latent	Early latent : positive serology with a negative test within 1 year, or recent contact. Late latent/unknown duration : all others	Seroreactive, no clinical disease
Tertiary	Neurosyphilis 5–12 y; spinal cord and brain after 10–20 y; cardiovascular 15–30 y	About a third of late latent cases: neurosyphilis, cardiovascular (any large vessel), gummatous. Tabes dorsalis (cord); general paralysis of the insane (brain). Rare outside developing countries. Neurosyphilis can occur at any stage, including secondary; meningovascular syphilis in tertiary

Diagnosis of Syphilis

- **Darkfield examination**: *T. pallidum* cannot be cultured in vitro; **darkfield of lesion exudate or tissue is the definitive test for early syphilis**, but cumbersome and rarely used. No commercial direct-detection NAAT exists.
- **Serology — presumptive diagnosis needs two tests, one nontreponemal and one treponemal**. Either alone gives false negatives in primary syphilis and false positives in the uninfected.

NONTREPONEMAL VS TREPONEMAL TESTS

	Nontreponemal (NTT)	Treponemal (TT)
Examples	Venereal Disease Research Laboratory (VDRL), rapid plasma reagin (RPR)	Fluorescent treponemal antibody absorption (FTA-ABS), <i>T. pallidum</i> particle agglutination (TP-PA), enzyme immunoassay (EIA), chemiluminescence immunoassay (CIA), immunoblot, rapid assays
Detects	Antibodies to lipid bound to the treponeme	Antibodies to treponemal antigens
Turns positive	10–15 days after chancre onset	—
False positives	Common: other infections including HIV, autoimmune disease, immunisation, pregnancy, drug use, old age	Uncommon: collagen disease, lupus, other infections
After treatment	Titre falls , may become non-reactive — tracks activity, so used to monitor treatment ; persistent low titre = serofast	Usually positive for life ; 15–25% treated in the primary stage revert after 2–3 years

- **Clinically significant change = fourfold (two dilutions)**, e.g. 1:16 → 1:4 or 1:8 → 1:32. **Use the same NTT for serial testing.**
- **Every reactive NTT needs a confirmatory TT.**
- **Rapid treponemal tests** (EIA or CIA, FDA-approved) give results in 5–10 minutes but **cannot distinguish active from treated syphilis**.

Reverse screening algorithm:

1. Rapid TT screen → if positive, reflex NTT.
2. NTT negative → second TT, preferably on a different antigen.

3. Second TT positive with previous treatment → no further action unless sexual history suggests re-exposure (then repeat NTT in 2–4 weeks).
4. Second TT positive without previous treatment → **treat as latent syphilis**.
5. Second TT negative with low risk → no further evaluation.

Treatment for Syphilis

- **Parenteral (intramuscular) benzathine penicillin G is preferred for all stages.** Penicillin-allergic patients whose adherence or follow-up is doubtful → **desensitise and give benzathine penicillin. Never combine different penicillin G preparations.**

TABLE 29.7 TREATMENT OF SYPHILIS

Stage	Penicillin treatment	Penicillin-allergic
Primary, secondary, early latent (no neurological involvement)	Benzathine penicillin G 2.4 million units (MU) IM, single dose	Doxycycline 100 mg twice daily 14 days; or tetracycline 500 mg four times daily 14 days
Late latent or unknown duration (no neurological involvement)	Benzathine penicillin G 2.4 MU IM weekly for 3 weeks	Doxycycline 100 mg twice daily 28 days; or tetracycline 500 mg four times daily 28 days
Tertiary with normal cerebrospinal fluid (CSF)	Benzathine penicillin G 2.4 MU IM weekly for 3 weeks (7.2 MU total)	Consult infectious disease specialist
Neurosyphilis	Aqueous crystalline penicillin G 18–24 MU/day intravenously (IV) — 3–4 MU every 4 h or continuous infusion — 10–14 days. <i>Alternative:</i> procaine penicillin 2.4 MU IM daily + probenecid 500 mg orally four times daily, both 10–14 days	—

Data from Workowski et al., CDC 2021.

- **Pattern: treatment length tracks disease age** — one dose for early disease, three weekly doses for late disease, IV therapy for the CNS; the allergic alternative likewise doubles from 14 to 28 days.
- **Trap:** benzathine penicillin is **IM only**; neurosyphilis requires **aqueous crystalline penicillin IV**.
- **Jarisch–Herxheimer reaction:** acute fever, headache and myalgia **within 24 hours** of starting treatment, commonest in **early syphilis** (high treponemal burden). **It is a reaction to treponemal killing, not penicillin allergy**; warn patients; treat supportively with non-steroidal anti-inflammatory drugs (NSAIDs).
- **European 2020 guidance:** similar first line; **ceftriaxone 1 g IV daily for 10 days** is an option for early syphilis in patients with a **bleeding disorder**. CDC discusses but does not recommend ceftriaxone (optimal dose and duration unclear).
- **Follow-up:** clinical and serological review at **6 and 12 months**, plus **24 months for latent syphilis**. Persistent or recurrent signs, or a **sustained fourfold NTT rise for >2 weeks** → treatment failure or reinfection → retreat with benzathine penicillin G 2.4 MU IM weekly for 3 weeks (unless neurosyphilis), re-evaluate for HIV, consider CSF examination.

Chancroid

- **Organism and spread:** gram-negative bacillus *H. ducreyi*, sexually transmitted; **microabrasions are thought necessary** for entry. In the US, rare sporadic cases; **endemic in North India, the Caribbean and Malawi**. Commoner in lower socioeconomic groups, female commercial sex workers and their partners, and **uncircumcised males**. A risk factor for HIV transmission.

- **Course:** tender erythematous papules (prepuce and frenulum; vulva, cervix, perianal area) → pustules → rupture within days → superficial ulcers with **ragged, undermined edges**. **Incubation 3–7 days**. **Painful, usually unilateral inguinal lymphadenitis in about half** → **fluctuant buboes** that may rupture.
- **Definitive diagnosis** = *H. ducreyi* on special culture medium (not widely available); no FDA-cleared NAAT, though CLIA-verified laboratory-developed NAATs exist.
- **Clinical and surveillance diagnosis requires all four:** (1) ≥1 painful genital ulcer; (2) typical clinical picture, ulcer appearance and, if present, lymphadenopathy; (3) no *T. pallidum* on darkfield or NAAT of exudate or on syphilis serology ≥7–14 days after ulcer onset; (4) negative HSV-1/HSV-2 NAAT or culture of exudate.
- **Management:** HIV test (PrEP or more frequent testing if negative). **HIV-positive and uncircumcised patients respond less well** and may need longer treatment. **Fluctuant buboes → needle aspiration or incision and drainage.**

Granuloma Inguinale (Donovanosis)

- **Organism:** intracellular gram-negative *Klebsiella granulomatis* (formerly *Calymmatobacterium*). Rare in the US; reported from Papua New Guinea, South Africa, India and Brazil.
- **Lesion:** incubation about **50 days**; painless, slowly progressive ulcers on genitals or perineum **without regional lymphadenopathy** — despite the name, only 10% involve the inguinal region. **Highly vascular (beefy red), bleed easily**. Extragenital spread: mouth commonest; also pelvis, intra-abdominal organs, bone.
- **Diagnosis:** dark-staining **Donovan bodies on tissue crush preparation or biopsy**; no FDA-cleared molecular test. Test for HIV.
- **Biopsy if antibiotics fail** — penile squamous cell carcinoma mimics granuloma inguinale.

Lymphogranuloma Venereum

- **Serovars:** *C. trachomatis* is an obligate intracellular pathogen with 15 serovars in three groups — trachoma (A, B, Ba, C), anogenital infection (D–K), **LGV (L1, L2, L3)**.
- 1. **Primary:** self-limited, often unnoticed genital ulcer or papule 3–30 days after inoculation. Rectal inoculation → **proctitis:** rectal bleeding, pain, tenesmus, constipation, haemopurulent discharge. **Proctocolitis is now the commonest presentation.**
- 2. **Secondary (2–6 weeks later):** regional invasion with fever, chills, malaise, myalgia; penile, urethral or vulvar inoculation → **inguinal syndrome** — **unilateral, painful inguinal or femoral buboes**.
- 3. **Tertiary:** untreated → tissue destruction and scarring → lymphoedema and **genital elephantiasis**; rectal disease → perirectal abscess, fissures, strictures.
- **Inguinal nodes are commoner in males** because the cervix and vagina drain to **retroperitoneal**, not inguinal, nodes.
- **HIV:** LGV is not more frequent or more virulent.
- **Diagnosis:** clinical suspicion, epidemiology and **chlamydia NAAT at the symptomatic site**, excluding other causes; genital, rectal and node specimens by culture or NAAT — **NAAT preferred** (detects LGV and non-LGV strains). **All proctocolitis → rectal chlamydia NAAT**. Chlamydia serology is not used routinely.
- **Aftercare:** screen for HIV, gonorrhoea and syphilis; offer PrEP if HIV-negative.

TREATMENT — CHANCROID, DONOVANOSIS AND LGV

Disease	Recommended	Alternatives and notes
Chancroid	Any one: azithromycin 1 g orally once; ceftriaxone 250 mg IM once; ciprofloxacin 500 mg twice daily 3 days; erythromycin base 500 mg three times daily 7 days	European 2017: ciprofloxacin and erythromycin second line

Disease	Recommended	Alternatives and notes
Donovanosis	Azithromycin 1 g weekly or 500 mg daily for ≥ 3 weeks and until all lesions have healed	Alternatives in CDC guideline
LGV	Doxycycline 100 mg twice daily 21 days	Erythromycin base 500 mg four times daily 21 days; or azithromycin 1 g weekly 3 weeks (consider test of cure by NAAT 4 weeks after). A 7-day doxycycline course shows excellent cure, but CDC still advises 21 days

Molluscum Contagiosum

- **Organism:** benign epidermal eruption caused by molluscum contagiosum virus (MCV), a large double-stranded DNA virus; four types — **MCV-1 commonest, then MCV-2**, which is relatively common in immunocompromise and HIV.
- **As an STD:** young adults, increasing; genitals, pubis, lower abdomen, upper thighs, buttocks.
- **Lesion:** **smooth, firm, dome-shaped papules with central umbilication**, 2–5 mm, usually 1–30, asymptomatic. **Regress spontaneously in 6–12 months** without sequelae in the immunocompetent.
- **In HIV:** aggressive and widespread — **≥ 100 lesions**, confluent large lesions.
- **Diagnosis is clinical; biopsy atypical lesions** where malignancy must be excluded.
- **Management:** expectant management is acceptable in the immunocompetent; warn of auto-inoculation, avoid shaving and waxing, do not share towels, linen or clothes. Options: cryotherapy, curettage, light-emitting and pulsed-dye lasers; topical 0.5% podophyllotoxin and 5% imiquimod (**both avoided in pregnancy**) and other chemicals. **In HIV, the first treatment is highly active antiretroviral therapy (HAART)** — lesion number is inversely related to CD4 count and lesions may regress on HAART.

Concept — *ulcer diagnosis turns on pain and nodal pattern, but because lesions are so often atypical every ulcer receives the same test battery; treatment then follows the organism, with duration lengthening as infection becomes older or deeper* — syphilis by stage, LGV for 21 days, donovanosis until healed.

ANOGENITAL WARTS

Human Papillomavirus

- **Virology:** small non-enveloped double-stranded DNA viruses (Papillomaviridae); over 200 types, half genital. **HPV is the most prevalent STD in the world.**
- **High-risk types 16, 18, 31, 33, 35** cause most cervical, penile, vulvar, vaginal, anal and oropharyngeal cancers and precancers; **16 and 18 cause $>70\%$ of cervical cancer.**
- **HPV 6 and 11 are non-oncogenic and cause about 90% of anogenital warts.**
- **Burden:** infection is often unrecognised; most sexually active people are infected at least once. HPV causes roughly 57,000 cancers a year in females and 60,000 in males worldwide; in the US in 2009, about 35,000 new HPV-associated cancers and 355,000 new cases of warts.

Presentation and Diagnostic Considerations

- **Condylomata acuminata:** usually asymptomatic, painful or pruritic depending on size and site; flat, papular or pedunculated. Common at the vaginal introitus, under the foreskin (uncircumcised) and on the shaft (circumcised); also cervix, vagina, **urethra (meatal warts)**, perineum, perianal skin, anus, scrotum.
- **Vaccination has reduced wart incidence** in adolescents, young females and heterosexual males.

HPV-RELATED GENITAL LESIONS COMPARED

Lesion	HPV type	Features
Condyloma acuminatum	6 or 11	Benign wart
Bowenoid papulosis — now high-grade squamous intraepithelial lesion (HSIL)	16	Flat, often hyperpigmented papules, millimetres to centimetres, usually multiple; penis, near vulva, perianal
Giant condyloma acuminatum	6	Rare, aggressive verrucous carcinoma ; glans or prepuce of uncircumcised males, less often perianal skin or vulva; bland histology yet deep invasion , infrequent nodal metastasis

- **Trap:** giant condyloma is a carcinoma caused by **low-risk HPV-6**, unlike other HPV-induced genital cancers.
- **Diagnosis is clinical; HPV testing is not recommended** — it neither guides management nor confirms the diagnosis. External anal or perianal warts → inspect the anal canal (digital examination or anoscopy).
- **Biopsy if:** (1) diagnosis uncertain; (2) no response to standard therapy; (3) worsening during therapy; (4) immunocompromised; (5) **pigmented, indurated or fixed to underlying tissue**; (6) persistent ulceration or bleeding.

Treatment of Anogenital Warts

- **Goal:** removal of warts and relief of symptoms. Some resolve within a year, so **observation is acceptable**. Treatment may reduce infectivity, of unclear significance. Choice depends on size, number, site, preference, cost, adverse effects and provider experience.

ANOGENITAL WART TREATMENTS

Agent	Applied by	Regimen	Key points
Imiquimod 3.75% cream	Patient	Once nightly up to 8 weeks; wash off after 6–10 h	Induces interferon and cytokines → local inflammation (redness, irritation, induration, ulceration, vesicles), hypopigmentation; limited pregnancy data, animal data suggest low risk; weakens condoms and diaphragms
Imiquimod 5% cream	Patient	Nightly three times a week up to 16 weeks; wash off after 6–10 h	
Podofilox (podophyllotoxin) 0.5%	Patient	≤0.5 mL/day, twice daily for 3 days then 4 days off, up to 4 cycles; total area ≤10 cm²	Antimitotic → wart necrosis; provider demonstrates the first application; contraindicated in pregnancy
Sinecatechins 15% ointment	Patient	Three times daily up to 16 weeks; do not wash off	Green tea catechins (antioxidant, antiviral, immunostimulatory); avoid sexual contact during use; erythema, pruritus, burning, pain, ulceration, oedema, rash; not for HIV, immunocompromise, genital herpes or pregnancy ; weakens condoms and diaphragms
Cryotherapy (liquid nitrogen)	Provider	Repeat every 1–2 weeks	Cytolysis; local anaesthetic for large warts
Surgery: scissor excision, tangential shave, curettage, CO ₂ laser	Provider	—	Plastic surgical collaboration for large penile or groin-crease lesions; ventilated room with local exhaust ventilation

Agent	Applied by	Regimen	Key points
Trichloroacetic or bichloroacetic acid 80–90%	Provider	Weekly	Chemical coagulation of wart proteins; allow to dry before the patient stands; neutralise with soap and water or sodium bicarbonate if pain is intense

- **Pattern: pregnancy** — podofilox contraindicated, sinecatechins not recommended, imiquimod data limited; **condom weakening** — imiquimod and sinecatechins; **wash-off** — imiquimod yes, sinecatechins no.
- **Response:** most warts respond within 3 months; otherwise change modality and consider biopsy.
- **Counselling:** screen for other STDs; **HPV can still be transmitted after warts are treated**; condoms reduce but do not eliminate transmission (uncovered skin); vaccination prevents most warts in those not yet exposed.

Human Papillomavirus Vaccine

- **Prevention:** abstinence is most reliable; correct condom use and fewer partners reduce risk.

HPV VACCINES COMPARED

Vaccine	Types covered	Notes
Bivalent (Cervarix)	16, 18	—
Quadrivalent (Gardasil)	6, 11, 16, 18	Approved 2006 for females 9–26 y; 2009 for males 9–26 y
Nonavalent (Gardasil 9)	6, 11, 16, 18, 31, 33, 45, 52, 58	Only HPV vaccine available in the US since 2016

- **Type coverage:** 16 and 18 → about two-thirds of cervical cancer; 6 and 11 → 90% of warts; the five extra nonavalent types → a further 15% of cervical cancer.
- **Advisory Committee on Immunization Practices (ACIP):** nonavalent vaccine at **11–12 years (from 9)** and catch-up **through 26 years**.

DOSING SCHEDULE BY AGE AT FIRST DOSE

First dose	Schedule
Before 15th birthday	Two doses: second at 6–12 months (minimum 5 months); if given earlier, a third dose ≥12 weeks after the second and ≥5 months after the first
On or after 15th birthday, or immunocompromised	Three doses: 0, 1–2 and 6 months

- **Restrictions:** **not in pregnancy**; may be given regardless of prior HPV exposure but is **prophylactic, not therapeutic**; nonavalent vaccine is made in *Saccharomyces cerevisiae* → **contraindicated in yeast allergy**.
- **Safety:** large cohort data show no increase in autoimmune, neurological or venous thromboembolic events; vaccination is not associated with earlier sexual activity or riskier behaviour.

Concept — HPV disease divides by type: **6 and 11 make benign warts, 16 and 18 make dysplasia and cancer** — giant condyloma being the exception — and vaccine valency is simply a choice of which types to cover.

HUMAN IMMUNODEFICIENCY VIRUS AND ACQUIRED IMMUNODEFICIENCY SYNDROME

- **History and scale:** AIDS described in 1981; HIV identified in 1983. In 2021, 38.4 million people were living with HIV worldwide, with 1.5 million new cases and 650,000 AIDS-related deaths; about 1.2 million live with HIV in the US.

- **US new diagnoses (2020):** about 30,600, falling; **MSM about two-thirds**, heterosexual contact about a fifth, injection drug use 7%, both 4%; highest rate in Black/African Americans, then Hispanic/Latino and White populations.
- **Biology:** retrovirus infecting T cells and dendritic cells; spread by blood, semen, vaginal fluid and breast milk; immunosuppression from progressive **CD4 T-lymphocyte loss**.
- **HIV-1 causes most US infections. Suspect HIV-2** with an unusual presentation or endemic-area risk: residence there, a partner from there, a partner with HIV-2, or transfusion or non-sterile injection there.

STAGES OF HIV INFECTION

Stage	Features
1 — Acute	2–4 weeks after infection; flu-like; high viral load → high transmissibility; antibody tests often negative
2 — Clinical latency (chronic)	Slow replication for years to decades depending on treatment; usually asymptomatic
3 — AIDS	CD4 <200 cells/mm³ or an AIDS-defining condition

Diagnosis of Human Immunodeficiency Virus

- **Screen everyone aged 15–65 at least once** (opt-out: informed, may decline); screen **everyone seeking STD care; high-risk persons at least annually**.
- **Tests:** serological (antibodies to HIV-1 and HIV-2) and virological (antigen or RNA). Screening immunoassays are highly sensitive and specific and detect all HIV-1 subtypes; most detect HIV-2. **Rapid tests turn reactive later than laboratory assays → may miss recent infection.**
 1. Laboratory HIV-1/HIV-2 antigen–antibody combination immunoassay.
 2. Repeatedly reactive → HIV-1/HIV-2 antibody differentiation assay; positive confirms the diagnosis and identifies HIV-2 (different monitoring and treatment).
 3. Discordant (reactive immunoassay, negative differentiation assay — possible in very early infection) → **HIV RNA** to identify acute infection.
- **Partners:** encourage notification (health department services). Partners exposed to genital secretions or blood **within 72 hours → postexposure prophylaxis (PEP)** with combination antiretrovirals. HIV-negative persons at increased risk → consider PrEP, and PEP if the last exposure was ≤72 hours ago.

Decreasing the Transmission Risk

Male Circumcision

- **Three sub-Saharan African randomised controlled trials: circumcision cut heterosexual HIV acquisition by 50–60%**, with similar reductions in high-risk HPV and genital herpes.
- **Mechanisms:** abundant HIV target cells beneath the foreskin dermis; **poorly keratinised inner foreskin** prone to lacerations (entry sites); anaerobic preputial infection recruits **Langerhans cells**, which HIV targets.
- **Limits: no reduction in transmission from an HIV-positive male;** not shown to protect MSM, although limited data suggest benefit for exclusively insertive partners.
- **Policy:** the World Health Organization (WHO), the Joint United Nations Programme on HIV/AIDS (UNAIDS) and the US President's Emergency Plan for AIDS Relief (PEPFAR) recommend scaling up circumcision to prevent heterosexually acquired HIV.

Postexposure Prophylaxis

- **Landmark evidence:** a 1997 case-control study of health care workers (HCWs) with percutaneous exposure found **zidovudine prophylaxis reduced the odds of transmission by 81%**. No randomised trial of non-occupational PEP is feasible, so this remains the strongest evidence.
- **Non-occupational PEP (nPEP):** evaluate anyone presenting **<72 hours** after an exposure of substantial risk; if the exposed person's HIV test is negative or unavailable, **start without delay. Not recommended if risk is not substantial or presentation is >72 hours.**
- **Regimen: 28 days, three drugs** — tenofovir disoproxil fumarate (TDF) 300 mg + emtricitabine 200 mg daily **plus** raltegravir 400 mg twice daily **or** dolutegravir 50 mg daily. Test for HIV at 6 weeks, 3 months and 6 months.

FIGURE 29.12 ALGORITHM FOR NONOCCUPATIONAL POSTEXPOSURE HIV PROPHYLAXIS

Exposure risk	Time since exposure	Source	Decision
Substantial	≤72 h	Known HIV-positive	nPEP recommended
Substantial	≤72 h	HIV status unknown	Case-by-case
Substantial	≥73 h	Any	Not recommended
Negligible	Any	Any	Not recommended

From Updated guidelines for antiretroviral postexposure prophylaxis, MMWR 2016.

- **Substantial risk:** exposure of vagina, rectum, eye, mouth, other mucous membrane, non-intact skin or percutaneous contact **with** blood, semen, vaginal or rectal secretions, breast milk or any visibly bloody fluid, **when** the source is known to have HIV.
- **Negligible risk:** exposure of the same sites or intact skin to **urine, nasal secretions, saliva, sweat or tears not visibly contaminated with blood**, regardless of the source's status.
- **Occupational exposure:** only 58 confirmed US HCW transmissions. Risk: **percutaneous about 0.3%, mucous membrane about 0.09%, non-intact skin near zero. Three-drug PEP within 72 hours for all occupational exposures** to HIV-infected blood or fluid, similar to nPEP; a low source viral load does not change this.

Pre-exposure Prophylaxis

- **Regimens:** daily oral emtricitabine (F; FTC) 200 mg with **TDF 300 mg (males and females) or tenofovir alafenamide (TAF) 25 mg (males and transgender females).** F/TAF is not recommended for females or transgender males (no data).
- **Eligibility:** educate all sexually active adolescents and adults; recommended for persons **≥35 kg** whose sexual or injection behaviour increases risk.

Urologic Manifestations of HIV/AIDS

HIV and Other Sexually Transmitted Diseases

- **Test for HIV in anyone with, or at risk of, an STD;** HIV epidemiology mirrors that of other STDs.
- **Ulcerative STDs (herpes, syphilis, chancroid) increase both transmission and acquisition:** bleeding ulcers and HIV in exudate increase infectivity; in HIV-negative persons ulcers breach mucosa and recruit HIV-susceptible immune cells.
- **Coinfection:** HSV shedding increases in HIV; HSV-2 affects most HIV-positive patients (50–90%). **Syphilis raises viral load and lowers CD4 count → rapid progression.**

Nonulcerating Skin Infections

- **Perineal cellulitis** is commoner → empiric cover for **methicillin-resistant *Staphylococcus aureus* (MRSA).** Necrotising fasciitis is rare but commoner.

- **Tuberculosis (TB)** of scrotal skin or penis → sinuses, nodules or ulcers.
- **Warts** (HPV-6 and -11) in about 20%, sometimes extensive → giant condyloma. **Molluscum** in 5–18%; may not regress and can seed secondary bacterial infection.

Urinary Tract Infections

- **Risk rises as CD4 falls:** urinary tract infection (UTI) risk is increased at CD4 <500/mm³ and highest at <200/mm³ or with high viral load — bacteriuria in about 30% below 200, about 11% at 200–500, and none above 500.
- **Drivers:** immunosuppression chiefly; voiding dysfunction contributes.
- **Pathogens:** *E. coli*, *Enterobacter*, *Proteus*, *Klebsiella*, other gram-negatives; ***Pseudomonas aeruginosa* in up to 30%. *Salmonella* UTI → lifelong prophylaxis** (risk of fatal recurrence).
- **Urine culture before empiric therapy**, given resistance patterns.
- **Non-response after a few days, or CD4 <500 (especially <250) → screen for atypical and opportunistic organisms:** fungi (*Candida*, *Aspergillus*, *Blastomyces*, *Cryptococcus*, *Histoplasma*), *Cryptosporidium*, *Toxoplasma*, *Pneumocystis*, *M. tuberculosis*, cytomegalovirus (CMV), adenovirus. **Viral UTI: PCR diagnosis, cidofovir treatment.**

Upper Urinary Tract Infections

- **Pyelonephritis:** gram-negatives commonest; opportunists include *Pneumocystis jirovecii* (formerly *carinii*), mycobacteria, *Candida*, *Histoplasma*.
- **Renal mycobacterial infection** is found at autopsy in a notable minority of AIDS patients, mostly asymptomatic in life; ***M. tuberculosis* commonest**, then *M. avium* and *M. intracellulare*. Spread is **descending**, and infection may lie latent for years.
- **Renal TB diagnosis:** Ziehl–Neelsen stain of urine; acid-fast bacilli (AFB) culture of **3–6 early-morning urines, up to 6 weeks**; PCR (48 hours, not widely available).
- **Iliopsoas abscess:** about 10% tuberculous, and about a quarter of these HIV-associated.
- **Drug interaction:** rifampicin induces cytochrome P450 → **lowers protease inhibitor (PI) and non-nucleoside reverse transcriptase inhibitor levels → monitor.**
- **Annual urine AFB culture for up to 10 years after treated renal TB** (relapse risk).

Prostatitis

- **Frequency:** uncommon generally (1–2%), commoner with HIV and more so with AIDS (pre-HAART figures). Symptoms: chronic pelvic or perineal pain, dysuria, urgency, frequency, other lower urinary tract symptoms (LUTS).
- **Organisms:** *E. coli* usually; also *S. aureus*, *Klebsiella pneumoniae*, *P. aeruginosa*, *Serratia marcescens*, *Salmonella typhi*, *M. tuberculosis*, *M. avium*, *M. intracellulare*. **Fungi especially with CD4 <200:** *Candida albicans*, *Aspergillus fumigatus*, *Cryptococcus neoformans*, *Histoplasma capsulatum*.
- **Course:** **higher risk of prostatic abscess and urosepsis** (atypical organisms); prolonged antimicrobial or antifungal therapy; **relapse in up to 70%** despite it.
- **Cryptococcal prostatitis** persists after treatment in up to 29% and is a **reservoir for relapsing meningitis → long-term or lifelong antifungals** depending on immune status.

Testis, Epididymis, and Seminal Vesicles

- **Epididymo-orchitis** is common, recurrent and chronic; persistent inflammation alters epididymal tubules and impairs clearance of secretions.
- **Reservoir:** at autopsy a quarter to over a third of men dying of opportunistic infection carry the **same organism in the testis** — gonococcus, *Salmonella*, CMV, *M. avium*, *M. intracellulare*, *Toxoplasma*,

Histoplasma, *Candida*. Treat with initial antibiotics then **maintenance suppression, especially for *Salmonella***.

- **Testicular atrophy** is a frequent autopsy finding: fever, chronic illness, opportunistic infection, cachexia, cytokine activation, direct HIV cytotoxicity. **Body mass index (BMI) <20 → 3.5-fold risk; CMV end-organ disease → fivefold.**

Nephrolithiasis

- **Indinavir (a PI) is the classic culprit** — stones in up to 22% of treated patients; autopsy stones in up to 40% of HIV patients. Risk factors: hepatitis coinfection, PI exposure.
- **Mechanism:** 20% of indinavir is excreted unchanged within 24 hours; it is **soluble at pH <5 and precipitates in alkaline urine → radiolucent stones**. Indinavir is present in only 29% of calculi; the rest are calcium oxalate, ammonium acid urate and uric acid. Newer PIs (lopinavir, atazanavir, amprenavir, nelfinavir) cause stones far less often.
- **Other contributors:** malnutrition, diarrhoea, dehydration, urinary acidification, hypocitraturia, hyperuricosuria from cell lysis after chemotherapy for AIDS-associated lymphoma.
- **Work-up:** complete metabolic evaluation — stone analysis, 24-hour urine, serum creatinine, uric acid, calcium.
- **First line for PI stones, if conservative management is possible: stop the drug and hydrate** (successful in up to 70%).

Renal Dysfunction

- **Still a leading cause of death despite HAART;** arises from infection and from treatment.
- **Causes:** volume depletion (diarrhoea, vomiting), sepsis, metabolic dysfunction, nephrotoxins, **HIV-associated nephropathy (HIVAN)**, ureteric obstruction (stones, clot, retroperitoneal lymphoma, fungal collections, fibrosis).
- **Nephrotoxic drugs:** antibiotics; antifungals (amphotericin B, pentamidine); antivirals; antiretrovirals (indinavir, atazanavir, abacavir, tenofovir).
- **HAART era:** longer survival → more chronic kidney disease, hypertension and diabetes (**fourfold higher diabetes incidence on HAART**); untreated HIV raises acute kidney injury risk.
- **HIVAN:** raised creatinine with **nephrotic-range proteinuria, often without oedema or hypertension**; typically **CD4 <200 and viral load >1000 copies/mL**; **about 12 times commoner in Black than White patients** — third leading cause of end-stage renal disease (ESRD) in Black patients aged 20–64. **Biopsy: collapsing focal segmental glomerulosclerosis**, tubular and visceral epithelial proliferation, tubular microcysts, oedema, interstitial fibrosis, leucocytic infiltrate. **Dialysis within about 10 months**; treat with HAART and angiotensin-converting enzyme (ACE) inhibitors; **1-year mortality about 50%.**

Voiding Dysfunction

- **LUTS worsen as disease progresses:** HIV is an independent risk factor for bothersome LUTS in MSM, and prior AIDS for severe LUTS.
- **Acute HIV: urinary retention and sacral sensory loss may herald seroconversion.**
- **Later causes:** recurrent or chronic UTI; CNS disease (HIV encephalitis, cerebral toxoplasmosis, demyelination, CMV polyradiculopathy, neoplasm, HIV dementia); peripheral neuropathy; HAART and other drugs.
- **CMV:** reactivates in about 40%, risking bladder damage and life-threatening infection. **Combined bladder and bowel dysfunction with back or sciatic pain → suspect CMV polyradiculopathy → lumbar puncture;** prompt treatment may help.

Erectile Dysfunction

- **Prevalence:** erectile dysfunction (ED) is commoner in HIV-infected males in every age decade; **HIV infection was the strongest independent predictor** on multivariate analysis, confirmed by meta-analysis. About half of HIV-positive men under 50 are affected.
- **Causes:** neurological, vascular and hormonal comorbidity, psychological stress, depression, hypogonadism, age, BMI. **Antiretrovirals, especially PIs** (indinavir, ritonavir), worsen ED: weight gain, dyslipidaemia, hyperglycaemia and lower testosterone → endothelial dysfunction.
- **Other dysfunction:** ejaculatory disorders commoner; **libido lower with HIV and lowest on HAART.**
- **Interaction:** PIs inhibit CYP3A → **sildenafil levels rise 3–10-fold → start at a low dose.**

Malignancy

- **Excess cancer** versus age-matched controls: reduced immune surveillance, viral protein effects, cytokine dysregulation, and behaviour (smoking two to three times commoner).

AIDS-DEFINING VS NON-AIDS-DEFINING CANCERS

	AIDS-defining cancers (ADCs)	Non-AIDS-defining cancers
Examples	Kaposi sarcoma (KS; human herpesvirus 8, KS-associated herpesvirus), non-Hodgkin lymphoma (NHL), invasive cervical carcinoma	Prostate; carcinoma of anus, lung, breast, skin, conjunctiva, head and neck, liver, testis; Hodgkin lymphoma; plasma-cell neoplasia; myeloma; leukaemia; melanoma; leiomyosarcoma
Effect of HAART	KS and NHL reduced; cervical and anal cancer not	Threefold increase; now about 70% of cancers in treated HIV (20% pre-HAART)

Concept — *HIV urology is the urology of a falling CD4 count: organisms become opportunistic, infections become chronic with the genitourinary tract as a reservoir, and in the HAART era the burden shifts toward drug toxicity and diseases of ageing — stones, nephrotoxicity, ED, non-AIDS cancers.*

ECTOPARASITIC INFECTIONS

Pediculosis Pubis (*Phthirus pubis*): Pubic or Crab Louse

- **Transmission:** sexual or close body contact, less often fomites; infests **terminal hairs** of the pubic and perianal areas. Adult lifespan <1 month; eggs hatch in 1 week; incubation usually up to a month.
- **Condoms do not prevent transmission. Pubic lice in children do not prove sexual contact.** Evaluate for other STDs including HIV.
- **Clinical:** genital itch; nits or lice attached to hairs. Symptoms 2–6 weeks after first exposure, sooner on re-exposure. **Diagnosis needs live lice or viable eggs** — empty shells persist for months.
- **Aftercare:** no fumigation needed; treat partners from the previous month; review at 1 week and retreat if lice or eggs remain; switch to an alternative regimen if there is no response.

Scabies

- **Transmission:** *Sarcoptes scabiei* by skin-to-skin contact, including intercourse, less often fomites. The female burrows into the epidermis and lays eggs that mature in 2 weeks; **mites die off the host within 24–36 hours.**
- **Symptoms are immunological:** type IV hypersensitivity to mites and products → symptoms **3–6 weeks after first infestation but 1–3 days after reinfestation.**
- **Presentation:** rash and itch, **worse at night**; burrows as tiny raised serpiginous lines. Scratching → *S. aureus* or β -haemolytic streptococcal infection → **post-streptococcal glomerulonephritis.**

- **Crusted (Norwegian) scabies:** heavy mite burden forming crusts; immunodeficient, debilitated or malnourished patients.
- **Diagnosis:** itch, findings and exposure history; **definitive — skin scrapings showing mites, eggs or faecal pellets.** Screen sexually active patients for STDs including HIV.
- **Contacts and outbreaks:** examine sexual, close and household contacts from the previous month and treat the infested; epidemics in nursing homes, hospitals and residential facilities may use ivermectin with specialist input.

ECTOPARASITE TREATMENT COMPARED

Condition	Recommended	Alternatives and cautions
Pubic lice	Permethrin 1% cream rinse, washed off after 10 min; or pyrethrins with piperonyl butoxide (not near the eyes)	Resistance increasing → malathion 0.5% lotion (off after 8–12 h) or ivermectin 250 µg/kg orally, repeated in 7–14 days
Scabies	Permethrin 5% cream; ivermectin 200 µg/kg orally repeated at 2 weeks; or ivermectin 1% lotion (repeat at 1 week if symptoms persist). Topicals applied at night to all skin — scalp, groin, navel, genitalia, finger and toe webs, under nail ends — for 8–14 h	Lindane 1% only if ≥10 years and other therapies not tolerated or failed (toxicity)
Crusted scabies	Topical scabicide (benzyl benzoate 25% or permethrin 5%) full-body daily for 7 days, then twice weekly until cured, plus ivermectin 200 µg/kg on days 1, 2, 8, 9, 15 (and 22, 29 if severe)	Never lindane; standard regimens often fail

- **Shared environmental measures:** machine-wash bedding and clothing at 50 °C, dry-clean, or seal in a plastic bag for 3 days.
- **Pattern:** lice → **permethrin 1%**, scabies → **permethrin 5%**; ivermectin 250 µg/kg for lice versus 200 µg/kg for scabies.

VAGINITIS

- **Symptoms:** discharge, odour, itch, irritation, burning; most females have at least one episode.
- **Commonest causes:** **bacterial vaginosis (BV), vulvovaginal candidiasis, trichomoniasis;** BV and trichomoniasis are sexually transmitted. Diagnosis by **pH, potassium hydroxide (KOH) preparation and microscopy of fresh discharge.**

TABLE 29.9 DIFFERENTIAL DIAGNOSIS OF VAGINITIS IN FEMALES

	Normal	Candidiasis	Trichomoniasis	Bacterial vaginosis
Discharge	White, thick, smooth	White, thick, curd-like	Frothy or purulent	Thin, white, homogeneous
pH	≤4.5	≤4.5	>4.5	>4.5
WBC	Absent	Absent	Present	Absent
Microscopy	Lactobacilli	Mycelia	Motile trichomonads	Clue cells; paucity of lactobacilli
Amine odour	—	—	Present	Present (fishy)
Symptoms and signs	None	Vulvar pruritus, external dysuria	Vulvar erythema and oedema; punctate strawberry cervix	Fishy odour, increased discharge

- **What the table tests:** pH separates candidiasis (normal pH) from the two sexually transmitted causes (raised pH); WBCs then separate trichomoniasis from BV.

Bacterial Vaginosis

- **Polymicrobial replacement:** loss of hydrogen peroxide–producing *Lactobacillus* with overgrowth of anaerobes (*Prevotella*, *Mobiluncus*), *Gardnerella vaginalis*, *Ureaplasma*, *Mycoplasma* and fastidious anaerobes.
- **Commonest diagnosis in symptomatic females, yet most cases are asymptomatic.** Associations: multiple partners, douching, no condoms, lactobacillus deficiency.
- **Consequences:** increased risk of STDs, post-gynaecological-surgery complications, pregnancy complications and recurrence; **increases HIV transmission to the male partner.**
- **Gold standard: Gram stain** quantifying lactobacilli against gram-negative and gram-variable rods and cocci and curved gram-negative rods.
- **Clinical diagnosis (Amsel) — three of four:** (1) thin, white, homogeneous discharge coating the walls; (2) **clue cells**; (3) **pH >4.5**; (4) fishy odour before or after 10% KOH. BV NAATs only in symptomatic persons.
- **Treat symptomatic patients:** metronidazole 500 mg orally twice daily 7 days; or metronidazole 0.75% gel 5 g intravaginally daily 5 days; or clindamycin 2% cream 5 g intravaginally at bedtime 7 days (**weakens latex for 5 days**). Test for other STDs including HIV. **Routine partner treatment is not recommended.**

Trichomoniasis

- **Most prevalent non-viral sexually transmitted infection in the US** (about 3.7 million); protozoan *T. vaginalis*. Mostly asymptomatic; otherwise diffuse, malodorous, yellow-green discharge with or without vulvar irritation. **1.5-fold HIV acquisition**, preterm birth, other pregnancy complications.
- **Testing: NAAT is highly sensitive** (vaginal, endocervical or urine specimens); culture is less sensitive than NAAT; wet mount is cheap but sensitivity is low and falls within an hour — **read immediately.**
- **Treatment — females: metronidazole 500 mg twice daily for 7 days** (halves positive retests at 1 month compared with the old 2 g single dose) or tinidazole 2 g once. **Males: metronidazole 2 g once.**
- **Aftercare: no alcohol for 24 h after metronidazole, 72 h after tinidazole; retest all sexually active females within 3 months** (high reinfection); refer all partners for presumptive treatment.

Concept — the two sexually transmitted causes of vaginitis share a raised pH and differ in inflammation, and the partner rule follows transmission: **trichomoniasis — treat all partners; BV — routine partner treatment is not recommended.**

SEXUAL ASSAULT, ABUSE, AND SEXUALLY TRANSMITTED DISEASES

Diagnostic Tests

- **NAAT for chlamydia and gonorrhoea at sites of penetration or attempted penetration; MSM are offered NAAT regardless of the site of contact; females are tested for *T. vaginalis* by NAAT; everyone is screened for HIV, HBV and syphilis.**

Presumptive Treatment for Survivors of Sexual Assault

PRESUMPTIVE ANTIBIOTICS — ADOLESCENTS AND ADULTS

Survivor	Regimen
Females	Ceftriaxone 500 mg once (1 g if ≥ 150 kg) + doxycycline 100 mg twice daily 7 days + metronidazole 500 mg twice daily 7 days
Males	Ceftriaxone 500 mg once (1 g if ≥ 150 kg) + doxycycline 100 mg twice daily 7 days

- **Pregnancy:** consider emergency contraception.

- **HBV:** assailant HBsAg-positive → unvaccinated survivor receives **hepatitis B immune globulin (HBIG) and HBV vaccination**; assailant status unknown → HBV vaccine.
- **HPV:** unvaccinated or incompletely vaccinated survivors aged 9–26 → vaccinate at the first examination, then at standard intervals.
- **HIV:** test the assailant if possible; **nPEP within 72 hours according to risk assessment** (Figure 29.12).

Concept — *the presumptive regimen is the sum of the organisms targeted: ceftriaxone for gonorrhoea, doxycycline for chlamydia, metronidazole for trichomonas — hence metronidazole for females only.*

KEY POINTS

- Patients with urethritis need treatment for both gonorrhoea and chlamydia. Because of higher sensitivity, **NAATs are the preferred method** for detecting *N. gonorrhoeae* and *C. trachomatis*; urethral swab is no longer indicated.
- Males with **recurrent urethritis should be tested for *M. genitalium***.
- Most genital ulcers in the US are **herpes (most common) or syphilis**. Chancroid occurs in parts of the US; granuloma inguinale usually does not. **LGV is increasing in MSM**.
- **HPV vaccines** are recommended for males and females through 26 years, preferably before sexual debut.
- **HIV testing** is recommended for anyone with, or at risk of, an STD.
- HIV is becoming a **chronic disease**; many associated problems now stem from ageing and chronic illness rather than immunosuppression.